

Riddhiman Raguraman *Software Engineer*

raguramanriddhiman@gmail.com · +1 773 751 9601 · Chicago, IL · [Portfolio](#) · [LinkedIn](#) · [GitHub](#)

C++ Software Engineer specializing in 3D programming, real-time rendering systems, and GPU performance optimization. Experienced in DirectX 11, HLSL shader authoring, and hardware-aware rendering architecture. Currently pursuing a Master's degree in Game Programming.

Education

MS in Game Programming, [DePaul University](#) Sep 2024 – Present

Chicago, IL | CGPA: 3.16 / 4.00

Relevant Course work: GPU Architecture, CUDA Programming, Rendering Pipeline, Parallel Programming, Multithreaded Programming, Real Time System Architecture.

Main Project: [Game Engine Development](#) C++ / DirectX 11 Apr 2025 – Present

- Built a DirectX 11 rendering engine across a 2,120-file codebase: SIMD-accelerated math library (vectors/matrices/quaternions) with ~4× SSE speedup, HLSL vertex/pixel/compute shaders, and a tree-based scene graph with hierarchical transforms and keyframe animation.
- Implemented view-frustum culling (AABB/OBB/sphere) and a job-based architecture decoupling update, physics, and render; wrote GPU compute shaders in HLSL to offload skeletal animation skinning from CPU to GPU; offline glTF → Protocol Buffers asset pipeline with a native Windows virtual file system.

BTech Computer Science, [VIT Chennai](#)

2020 – 2024

Chennai, India | CGPA: 8.10/10.00 (Equiv. 3.24/4.00)

Relevant Course work: Data Structures and Algorithms, Computer Architecture, Networking and Distributed systems, Operating Systems, and Game Development.

Final Project: [Adaptive NPC in Serious Games](#) Unity / Python / LLM API | Aug 2023 – Apr 2024

- Developed an Android math-learning game in Unity featuring an AI-powered NPC teacher built on a Neuro-Symbolic model (neural network + symbolic reasoning) to guide students without giving away direct answers; integrated voice input/output via Unity WebSocket and a custom LLM API.
- Conducted a user study across 39 participants (ages <18 to 35+); 94.8% found the AI resolved their doubts and 92% rated it easy to use, with 79% completing at least one full level. Designed and implemented a Neuro-Symbolic AI architecture connecting a fine-tuned LLM to Unity via WebSocket.

Certifications

Master Certificate in Unreal AR/VR

Jun 2023 – Aug 2024

Monolith Research and Training Labs, Chennai

Introduction to Concurrent Programming with GPUs

Jan - Apr 2026

Coursera, Johns Hopkins University

Skills

C++: SIMD, STL, Multithreading, Cache Optimization

Graphics: DirectX 11, HLSL, Compute Shaders, XAudio2

CUDA: Warps, Grid, Atomic instructions

Engines: Unreal Engine, Unity, Blender

Languages: C, C#, Java, Python, Assembly, Lua, Bash

Platforms: Windows, Linux

Networking: Distributed Systems, Sockets, gRPC, ZeroMQ

Build: CMake, Premake, Visual Studio ID

Professional Experience

Game Developer Intern, [Geons Logix](#), Madurai, India

May 2023 – Sep 2023

Developed full gameplay in C# for a Unity 2D educational mobile game "Cloud's Rescue" — core mechanics, interaction systems, and dynamic level progression for 10th-grade students. Integrated custom Blender assets and used ShaderLab for visual effects. For more information [< Click – Here >](#)

Other Academic Projects

Additional information, pictures and videos of these projects can be found in <https://riddhimanraguraman.github.io/PortfolioWebsite/>

[Space Invaders](#) Windows C# Project | Jan 2025 – Apr 2025

Built a retro Space Invaders game with an engine-style architecture (scenes, object lifecycle, timers). Implemented Scene/State flow with explicit Enter/Leave transitions. Designed a Composite GameObject hierarchy and a Visitor Pattern -based collision dispatch system. Scene Switching is done using State pattern. Used Observer + Command pattern to drive collision responses, scheduled gameplay events, animations, and scene switching. Applied Factory/Strategy/Proxy/Null Object patterns plus manager-style pooling to keep systems modular and reusable.

[Multi-threaded Maze Solver](#) C++ Multithreading Project | Dec 2025

Engineered a bidirectional DFS solver for 15K×15K and 20K×20K mazes using two concurrent threads exploring from start and end simultaneously, with branch skipping to reduce frontier size and path stitching on meet detection. Achieved ~5.4× speedup over single-threaded BFS (5.96s → 1.10s) using entirely lock-free synchronization — cell-level coordination via std::atomic_uint with fetch_or, acquire-release ordering for found signaling, and zero mutexes.

[Custom Malloc and Free](#) C++ Console Project | Dec 2024

Designed a custom memory allocator in C++ using a next-fit free-list strategy with explicit used/free node tracking; implemented automatic coalescing of adjacent free blocks on deallocation to minimize fragmentation. Benchmarked ~1.96× faster than MSVC's default allocator (333ms vs 653ms on Ryzen 9 5900HX); enforced allocator integrity via thorough assertions across all allocation and deallocation paths.

[Multi-threaded Audio Engine](#) C++ Multithreading Project | Oct 2025

Engineered a real-time audio system to seamlessly stitch and play 23 raw WAVE files without artifacts using Actor Model Multithreading. Designed a 25-thread architecture to decouple file I/O from audio rendering for glitch-free playback. Implemented a double-buffered Producer–Consumer model with circular queues for efficient inter-thread communication. Utilized mutexes and condition variables for safe thread synchronization (no spinlocks). Applied RAIL for robust resource and thread lifecycle management under real-time constraints.